

# Hugo Lebeau

\* 13 December 1997 (Amiens, France) • +33 624 687 985  
✉ hugo.lebeau@inria.fr • 🌐 hugolebeau.github.io • in hugo-lebeau  
🔗 HugoLebeau • ID 0009-0001-8317-3876 • atRlpOIAAAAJ

*Family and environmental considerations influence my career choices.*

## Experience

### Research

#### Inria, ENS Lyon (OCKHAM Team)

*Postdoctoral Researcher*

Theoretical study of compressive learning models.

**Lyon, France**

*Feb. 2025–Present*

#### Université Grenoble Alpes (LIG & GIPSA-lab)

*Doctoral Researcher*

Theoretical study of machine learning techniques with random matrix theory.

**Grenoble, France**

*Oct. 2021–Jan. 2025*

#### Université Grenoble Alpes (GIPSA-lab)

*Research Intern*

Analysis of online learning with random matrix theory.

**Grenoble, France**

*Apr. 2021–Sep. 2021*

#### CEA

*Research Intern*

Proximal algorithms to improve the quality of industrial neutron imaging.

**Saclay, France**

*Mar. 2020–Jul. 2020*

#### Politecnico di Milano

*Research Intern*

Statistical and numerical methods for functional data on complex multidimensional domains.

**Milan, Italy**

*May 2019–Jun. 2019*

### Industry

#### AXA Climate

*Data Scientist Intern*

Weather data modeling and risk assessment for parametric insurance pricing.

**Paris, France**

*Sep. 2019–Feb. 2020*

## Education

#### Université Grenoble Alpes (MIAI)

*PhD in Applied Mathematics*

“Teaching in Higher Education” track.

**Grenoble, France**

*2021–2024*

#### ENS Paris-Saclay

*Master MVA (MSc) — Mathematics and Machine Learning*

With honors of the jury.

**Gif-sur-Yvette, France**

*2020–2021*

#### ENSTA Paris

*Diplôme d'Ingénieur (MSc) — Applied Mathematics, Optimization and Data Science*

Ranked in the top 5% among 150 students.

**Palaiseau, France**

*2017–2021*

## PhD Thesis: *Random Matrix and Tensor Models for Large Data Processing*

**Manuscript:** <https://theses.hal.science/tel-05222408v2>

**Supervisors:** Romain Couillet, Florent Chatelain.

**Jury:** Philippe Loubaton, Rémi Bardenet, Mylène Maïda, Walid Hachem, Olivier Michel, Pierre Comon.

## Publications

### Journals

- H. Lebeau, F. Chatelain, and R. Couillet, *A Random Matrix Approach to Low-Multilinear-Rank Tensor Approximation*, Journal of Machine Learning Research (JMLR), vol. 26, no. 7, pp. 1–64, 2025.

○ H. Lebeau, F. Chatelain, and R. Couillet, *Asymptotic Gaussian Fluctuations of Eigenvectors in Spectral Clustering*, IEEE Signal Processing Letters, vol. 31, pp. 1920–1924, 2024.

**International Conferences**.....  
○ H. Lebeau, M. E. A. Seddik, and J. H. de M. Goulart, *Performance Gaps in Multi-view Clustering under the Nested Matrix-Tensor Model*, International Conference on Learning Representations (ICLR), 2024.  
○ H. Lebeau, R. Couillet, and F. Chatelain, *A Random Matrix Analysis of Data Stream Clustering: Coping With Limited Memory Resources*, International Conference on Machine Learning (ICML), 2022.

**National Conferences**.....  
○ H. Lebeau, *Performance of Rank-One Tensor Approximation on Incomplete Data*, Colloque GRETSI, 2025.  
○ H. Lebeau, R. Couillet, and F. Chatelain, *HOSVD Tronquée : Analyse d'une Approximation Tensorielle Rapide*, Colloque GRETSI, 2023  
○ H. Lebeau, R. Couillet, and F. Chatelain, *Une analyse par matrices aléatoires de l'apprentissage en ligne : traiter des grandes données avec des ressources mémoire limitées*, Colloque GRETSI, 2022.

Teaching

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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| <b>Random Matrix Theory and Machine Learning</b><br><i>Teaching Assistant, 9h/year</i><br>Introduction to the theory of large random matrices and their applications to machine learning with practical applications.                                                             | <b>ENS Paris-Saclay, Master MVA</b><br><i>Since spring 2022</i>      |
| <b>Statistics</b><br><i>Teaching Assistant, 24h/year</i><br>Advanced statistics topics around parametric estimation, maximum likelihood estimation, testing, regression, concentration of random variables.                                                                       | <b>ENS Lyon, M1</b><br><i>Spring 2025</i>                            |
| <b>Probability and Statistics (STA401)</b><br><i>Teaching Assistant, 18h/year</i><br>Basics of probability, standard probability laws, descriptive statistics, estimation, hypothesis testing.                                                                                    | <b>Université Grenoble Alpes, L2</b><br><i>Spring 2023, 2024</i>     |
| <b>Introduction to Artificial Intelligence (INF103)</b><br><i>Teaching Assistant, 18h/year</i><br>Introduction to basic concepts of machine learning: datasets, classifiers, training, performance evaluation, data processing.                                                   | <b>Université Grenoble Alpes, L1</b><br><i>Fall 2021, 2022, 2023</i> |
| <b>Introduction to Machine Learning</b><br><i>Teaching Assistant, 18h/year</i><br>Overview of the main tools in machine learning: model assessment, discriminant analysis, PCA, GLM and penalization, clustering with $k$ -means and EM, trees and random forests, deep learning. | <b>Grenoble INP, ENSE<sup>3</sup>, M2</b><br><i>Fall 2022</i>        |
| <b>Functional Programming</b><br><i>Teaching Assistant, 36h/year</i><br>Introduction to functional programming with OCAML.                                                                                                                                                        | <b>Université Grenoble Alpes, L1</b><br><i>Spring 2022</i>           |

Technical skills

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|--------------------------------------------------------------------------------|
| <b>Programming languages</b> .....                                             |
| <b>Proficient in:</b> Python (Numpy, Scipy, Scikit-learn, Pytorch, Matplotlib) |
| <b>Familiar with:</b> R, Julia, MATLAB, C, C++, OCaml                          |
| <b>Software</b> .....                                                          |
| LaTeX, Git                                                                     |

Languages

|                              |                              |                             |
|------------------------------|------------------------------|-----------------------------|
| <b>French:</b> Mother tongue | <b>English:</b> Professional | <b>German:</b> Intermediate |
|------------------------------|------------------------------|-----------------------------|

Miscellaneous

○ *Sports & Endurance:* Road and trail running, cycling, and hiking. Deeply interested in sports science, specifically physiology, nutrition, and training methodologies.  
○ *Performing Arts:* Theater enthusiast, practicing improvisational theater.  
○ *Intellectual Interests:* Keen interest in physics, biology, philosophy, and French literature.